

Supplementary Material for “Codesign of Contour-Conformal Flux Guides and Three-Arc Rotor Surfacing for Permanent Magnet Synchronous Machines”

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S1. STRUCTURAL ANALYSIS OF THE ROTOR

TABLE S1: Mechanical stress and deformation under different operating conditions (operating points of Fig. S1; field plots in Fig. S2)

Condition	Torque (Nm)	Speed (rpm)	Max. Stress (MPa)	Max. Deformation (mm)
Rated (Point 1)	72	4000	12.82	0.0038
Rated @ max speed (Point 2)	39.21	12000	108.78	0.02949
Peak (Point 3)	180	4000	17.26	0.0060
Peak @ max speed (Point 4)	46.21	12000	108.78	0.02950

Governing condition 12 krpm: 108.78 MPa against the 150 MPa stacked yield of Table S2, safety factor 1.4; deformation 0.0295 mm against the 0.5–1 mm air gap. Ansys Mechanical.

TABLE S2: Mechanical properties of the materials used in structural analysis

Material	Young's Modulus (MPa)	Yield Strength (MPa)	Poisson's Ratio
Steel	2×10^5	250	0.3
Steel with stacking	1.2×10^5	150	0.3

The laminated core is represented by the reduced stacked properties.

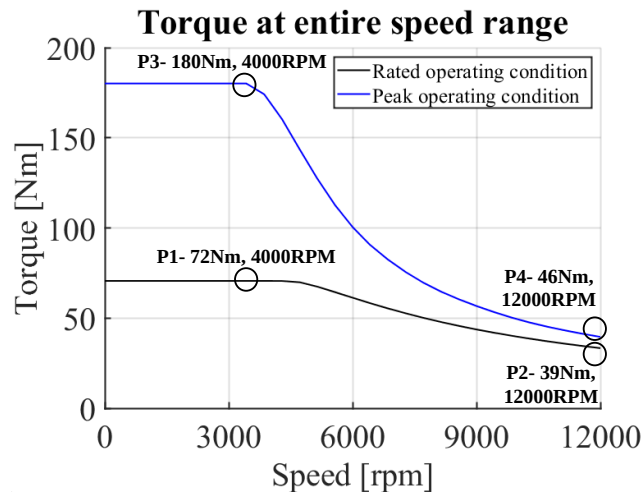
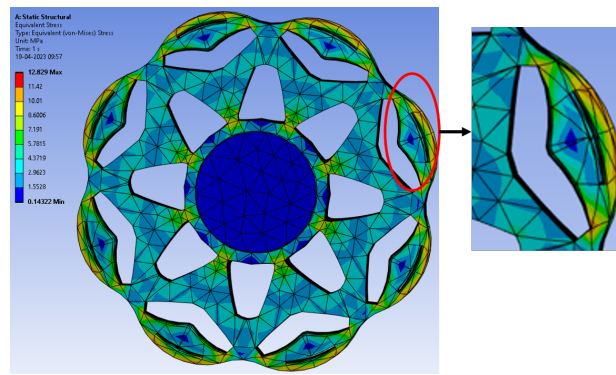
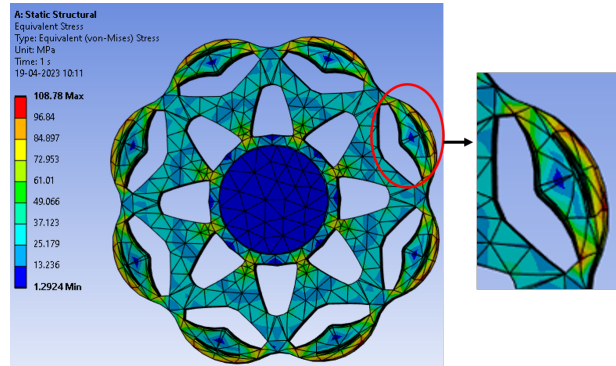


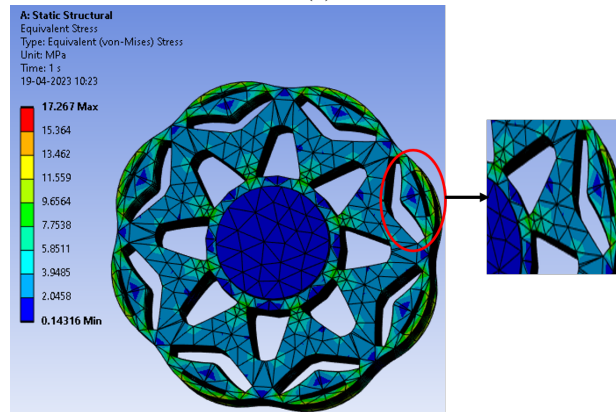
Fig. S1: Torque–speed operating points used for the structural analysis of Table S1, spanning rated and peak load at base and maximum speed.



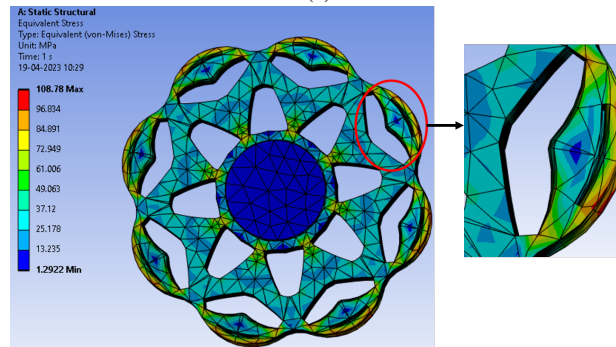
(a)



(b)



(c)



(d)

Fig. S2: Stress analysis of the rotor at the four operating conditions of Table S1: (a) Point 1, (b) Point 2, (c) Point 3, (d) Point 4. The clearances between the flux guide and the rotor surface, and between the ducts and the bore, are fixed from these results before the electromagnetic parametric study.

S2. ROTOR THERMAL BOUNDARY CONDITION

TABLE S3: Derivation of the convective film coefficient applied to the rotor boundary of the thermal model of Table VIII of the main paper. Duct values are evaluated at the maximum speed of 12 krpm.

Quantity	Expression	Value	Unit
Duct base	b	35.5	mm
Duct height	h_t	23.0	mm
Duct base radius	r_{base}	32.66	mm
Hydraulic diameter	$D_h = 4A/P$	17.45	mm
Centroid radius	$r_c = r_{\text{base}} + h_t/3$	40.33	mm
Angular velocity at 12 krpm	$\omega = 2\pi N/60$	1256.6	rad/s
Peripheral velocity	$U = \omega r_c$	50.7	m/s
Reynolds number	$Re = \rho U D_h / \mu$	5.2×10^4	–
Friction factor	$f = (0.79 \ln Re - 1.64)^{-2}$	0.021	–
Nusselt number, duct	$Nu_d = \frac{(f/8)(Re - 1000) Pr}{1 + 12.7\sqrt{f/8} (Pr^{2/3} - 1)}$	107	–
Film coefficient, duct	$h_d = Nu_d k / D_h$	172	W/(m ² K)
Taylor number, annulus	$Ta = \omega^2 r_i g^3 / \nu^2$	–	–
Nusselt number, annulus	$Nu_r = 0.128 Ta^{0.367}$	–	–
Film coefficient, annulus	$h_r = Nu_r k / g$	–	W/(m ² K)
Effective coefficient	$h_{\text{eff}} = \sum_i h_i A_i / \sum_i A_i$	Table VIII	W/(mm ² K)

Duct cross-section approximated as triangular of area A and wetted perimeter P . Air properties at the bulk condition

$T_\infty = 45^\circ\text{C}$: $\rho = 1.11 \text{ kg/m}^3$, $\mu = 1.9 \times 10^{-5} \text{ Pa}\cdot\text{s}$, $k = 0.028 \text{ W/(m}\cdot\text{K)}$, $Pr = 0.70$ (ref. [18] of the main paper).

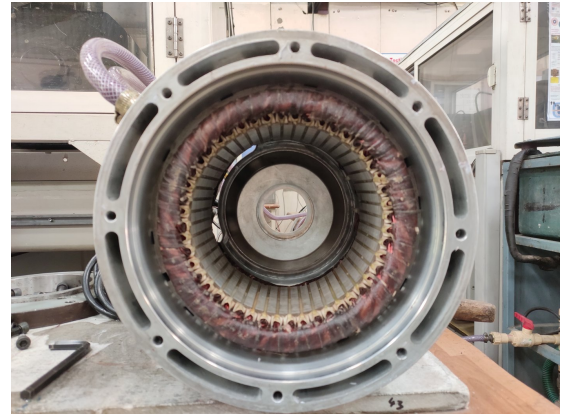
Nu_d from the Gnielinski correlation (ref. [19]); Nu_r from rotating-annulus Taylor–Couette convection (ref. [20]).

The Gnielinski correlation is written for the axial bulk velocity of a straight duct; the peripheral velocity at the duct centroid is substituted here, so h_d is a first-order design estimate for a rotating duct rather than a measured or CFD-resolved value, used only to set the rotor boundary condition. Annulus quantities depend on rotor radius r_i and air-gap length g ; the combined h_{eff} at each of the four operating points is listed in Table VIII of the main paper.

S3. PROTOTYPE LAMINATIONS AND WOUND STATOR



(a)



(b)

Fig. S3: Prototype developed from rotor Design E: (a) stator and rotor laminations, (b) housing with stator core and stator windings. The spline and arc geometry of Section II of the main paper was produced by commercial laser cutting of 0.5 mm laminations.

S4. ADDITIONAL FINITE-ELEMENT RESULTS

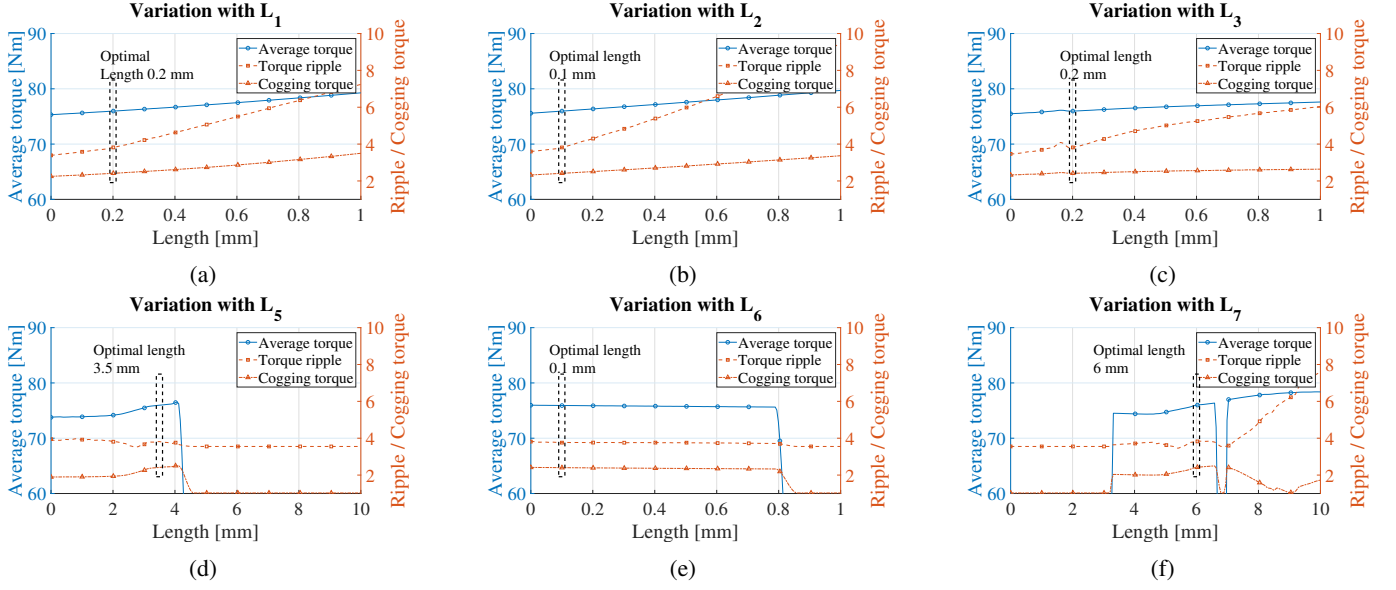


Fig. S4: Flux-guide parameter sweeps not reproduced in the main paper: (a) L_1 , (b) L_2 , (c) L_3 , (d) L_5 , (e) L_6 , (f) L_7 . Each length is swept individually with all other parameters at the values of Table II of the main paper; the selected value is marked on each panel. For L_1 – L_3 the average torque and T_{ripple} rise together, so the selected value lies at the low-ripple end of the range. For L_5 – L_7 the outputs are flat and then terminate, the termination marking the length beyond which the spline violates the clearance rules of Section II.

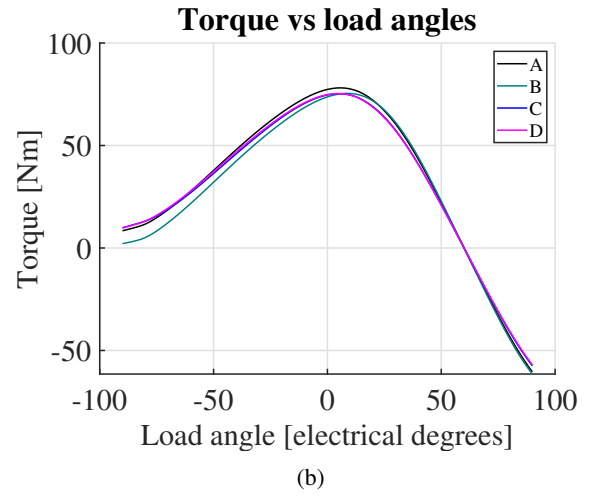
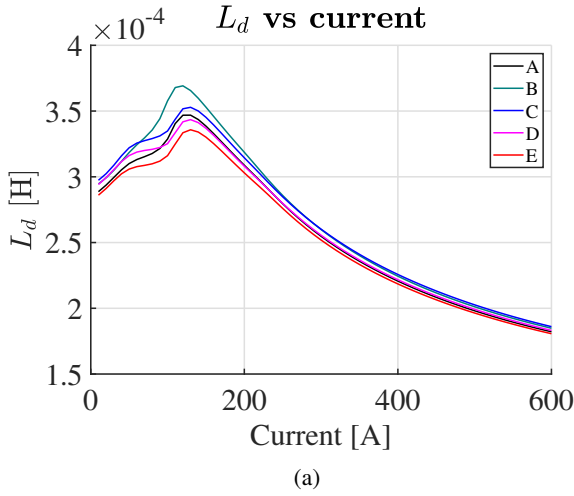


Fig. S5: Additional simulation results for the five rotor designs: (a) L_q against current, (b) torque against load angle at rated current, from which the operating load angle was selected. The direct-axis characteristic appears as Fig. 8(a) of the main paper.

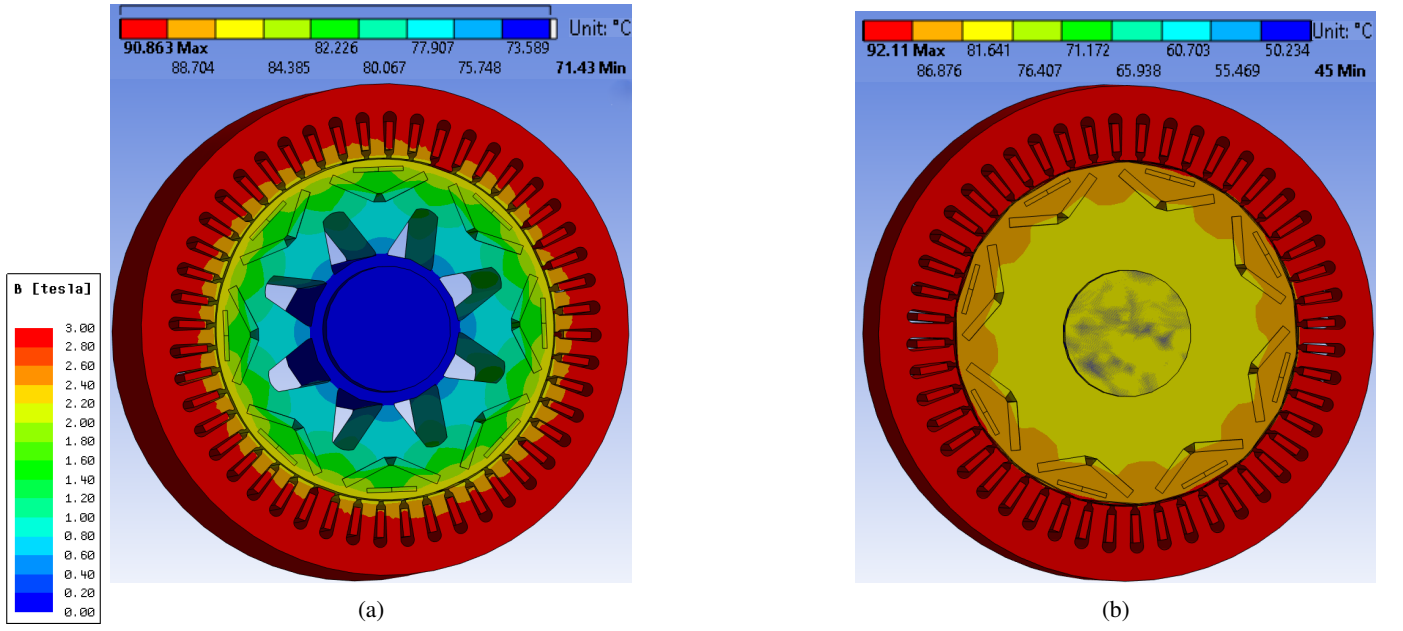


Fig. S6: Rotor temperature at the rated operating point, on a common scale: (a) with rotor ducts, (b) without rotor ducts. The peak-load maps appear as Fig. 9 of the main paper. At the rated point the ducts leave the hot spot essentially unchanged and reduce the rotor minimum temperature (Table VIII of the main paper); the hot-spot reduction appears only at the maximum-speed peak-load point.